Supplementary Material for: SPA-Bench: A Novel Video Benchmark for Proactive Physical Hazard Prediction in MLLMs

Anonymous Author(s)

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A Data Release and Ethical Considerations

Licensing and Access. To comply with YouTube's Terms of Service, we do not distribute raw video files. Instead, comprehensive annotations are released under the CC BY-NC 4.0 license for academic research. These include YouTube video IDs, clipping boundaries, cropping specifications, early warning timestamps (T_1 and T_2), and causal labels (H_1, H_2). We do not provide automated downloading tools; researchers must independently obtain the source videos using the provided IDs. Our scripts can then be applied to clip and crop the videos into the SPA-Bench format.

Ethical Considerations. All footage is sourced from publicly accessible YouTube videos. To maintain in-the-wild visual authenticity, no artificial privacy masking (e.g., face-blurring) is applied. Crucially, strict ethical filtering was enforced during manual review: videos depicting fatal incidents, extreme gore, or severe injuries were explicitly discarded.

B Evaluation Prompts

In this section, we present the prompts used to evaluate the model's inference results using `gemini-3-flash`. The prompts utilized for the model inference itself have been detailed in the main manuscript and are thus omitted here.

B.1 Prompts for Full-Video Evaluation

```
sys_prompt = """You are an objective expert judge evaluating
      a vision-language model's responses to sports accident
      videos.
You will be provided with the Ground Truth Hazard Category (
     Level 1) and Ground Truth Hazard Detail (Level 2).

Background:
- Level 1 factors belong to four macro categories: Personal
     Factors, Equipment and Facility Factors, Venue and
     Environmental Factors, Interpersonal Interaction
     Factors. A video may have one or multiple Level 1
     factors.
- Level 2 is the specific description of the direct cause of
      the accident.

Scoring Rubric for Level1, Level2, and Level3 responses (
     Score 0 or 1):
1. "mentioned_hazard": Score 1 if the response points out,
     describes, or acknowledges any danger, hazard, accident
     , or collision. Score 0 if it misses the accident
     completely.
2. "matches_level1": Score 1 ONLY IF the response addresses/
     covers ALL the factors listed in the Ground Truth Level
      1. If it only covers part of the Level 1 factors,
     score 0.
3. "matches_level2": Score 1 ONLY IF the response clearly
     identifies the Ground Truth Level 2 as the actual or
     primary cause/hazard.
   PENALTY RULE: If the model adopts a "shotgun approach" (
        merely listing the true Level 2 cause alongside
        several other incorrect, hypothetical guesses
        without prioritizing it), or if the description is
        too vague, score 0. The model must show certainty,
        not just lucky guessing.

CRITICAL SCORING LOGIC (Hierarchical Constraints):
- If `matches_level2` == 1, then automatically `
     matches_level1` = 1 and `mentioned_hazard` = 1.
- If `mentioned_hazard` == 0, then automatically `
     matches_level1` = 0 and `matches_level2` = 0.
- If the model correctly identifies the broad Level 1
     category but fails to specify the exact Level 2 cause (
     or guesses too many without focus), the score should be
      (1, 1, 0).

Output STRICTLY in this valid JSON format (replace the
     example values with your actual evaluated scores. Use 0
      or 1 for the metrics):
{
  "Level1": {"mentioned_hazard": 0/1, "matches_level1": 0/1,
        "matches_level2": 0/1},
  "Level2": {"mentioned_hazard": 0/1, "matches_level1": 0/1,
        "matches_level2": 0/1},
  "Level3": {"mentioned_hazard": 0/1, "matches_level1": 0/1,
        "matches_level2": 0/1}
}"""


user_prompt = f"""Ground Truth Hazard Category (Level 1): {
     hazard_l1}
Ground Truth Hazard Detail (Level 2): {hazard_l2}

[Level 1 Response]: "{responses.get('Level1', '')}"
[Level 2 Response]: "{responses.get('Level2', '')}"
[Level 3 Response]: "{responses.get('Level3', '')}"
"""
```

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B.2 Prompts for Temporal Window Evaluation

```
sys_prompt = """You are an objective expert judge evaluating
      a vision-language model's ability to anticipate and
     identify sports hazards across different time windows.
You will be provided with the Ground Truth Hazard Category (
     Level 1) and Ground Truth Hazard Detail (Level 2).

Background:
- Level 1 factors belong to four macro categories: Personal
     Factors, Equipment and Facility Factors, Venue and
     Environmental Factors, Interpersonal Interaction
     Factors.
- Level 2 is the specific description of the direct cause of
      the accident.

Scoring Rubric for Window_A, Window_B, Window_C (Score 0 or
     1):
1. "mentioned_hazard": Score 1 if the response confirms
     there is a danger/hazard occurring or about to occur.
     Score 0 if it claims there is no danger or fails to
     identify an impending accident.
2. "matches_level1": Score 1 ONLY IF the response addresses/
     covers ALL the factors listed in the Ground Truth Level
      1. If it only covers part of the Level 1 factors,
     score 0.
3. "matches_level2": Score 1 ONLY IF the response clearly
     identifies the Ground Truth Level 2 as the actual or
     primary cause/hazard.
   PENALTY RULE: If the model adopts a "shotgun approach" (
        merely listing the true Level 2 cause alongside
        several other incorrect, hypothetical guesses
        without prioritizing it), or if the description is
        too vague, score 0. The model must show certainty,
        not just lucky guessing.

CRITICAL SCORING LOGIC (Hierarchical Constraints):
- If `matches_level2` == 1, then automatically `
     matches_level1` = 1 and `mentioned_hazard` = 1.
- If `mentioned_hazard` == 0, then automatically `
     matches_level1` = 0 and `matches_level2` = 0.
- If the model correctly identifies the broad Level 1
     category but fails to specify the exact Level 2 cause (
     or guesses too many without focus), the score should be
      (1, 1, 0).

Output STRICTLY in this valid JSON format (replace the
     example values with your actual evaluated scores. Use 0
      or 1 for the metrics):
{
  "Window_A": {"mentioned_hazard": 0/1, "matches_level1":
       0/1, "matches_level2": 0/1},
  "Window_B": {"mentioned_hazard": 0/1, "matches_level1":
       0/1, "matches_level2": 0/1},
  "Window_C": {"mentioned_hazard": 0/1, "matches_level1":
       0/1, "matches_level2": 0/1}
}"""


user_prompt = f"""Ground Truth Hazard Category (Level 1): {
     hazard_l1}
Ground Truth Hazard Detail (Level 2): {hazard_l2}

[Window_A Response]: "{responses.get('Window_A', '')}"
[Window_B Response]: "{responses.get('Window_B', '')}"
[Window_C Response]: "{responses.get('Window_C', '')}"
"""
```

B.3 Prompts for the Diagnostic Analysis

```
sys_prompt = """You are an objective expert judge evaluating
      a vision-language model's ability to anticipate and
     identify sports hazards early on.

Scoring Rubric (Score 0 or 1):
- "mentioned_hazard": Score 1 if the response confirms there
      is a danger/hazard occurring or about to occur. Score
     0 if it claims there is no danger, talks about normal
     activities, or fails to identify an impending accident.

Output STRICTLY in this valid JSON format (replace the
     example values with your actual evaluated scores, using
      0 or 1):
{
  "Prompt1": {"mentioned_hazard": 0/1},
  "Prompt2": {"mentioned_hazard": 0/1}
}"""


user_prompt = f"""[Prompt1 Response]: "{responses.get('
     Prompt1', '')}"
[Prompt2 Response]: "{responses.get('Prompt2', '')}"
"""
```

C Video Search Keywords

In this section, we detail the keywords used for our video search on YouTube. Specifically, the search queries were constructed by combining keywords from the following two categories:

- **Sports Category Keywords:** "Basketball", "Soccer", "Skiing", "Parkour", "Gym", "Weightlifting", "Skateboarding", "Gymnastics", "Track and field", "Pole vault", "Trampolining", "Cycling", "Running", "Baseball", "Diving", "Sport".
- **Accident Description Keywords:** "injury", "accident", "broken leg", "fracture", "fail", "crash", "horrible", "snapped", "passed out", "gone wrong".

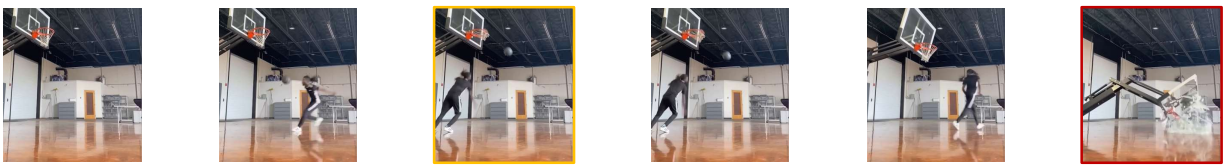
D Data Examples

In this section, we present several annotation examples.

Figures 1 to 14 provide 14 examples of accident video annotations. Figures 15 to 17 provide 3 examples of safety video annotations. Please note that the explanations of the timestamps shown in the figures are not included in the dataset.

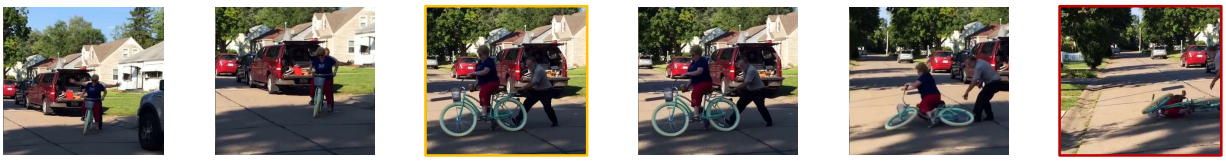
Environmental Setting: Standardized Sports Venues
Sports Category: Baseball
Macro Inducing Factors: Interpersonal Interaction Factors
Direct Cause Description: Collision while catching
Earliest Cue Moment (orange): We can see that the two players are about to collide
Most Obvious Moment (red): Clearly, the two players have collided and fallen to the ground

Figure 1: Accident video annotation example 1.



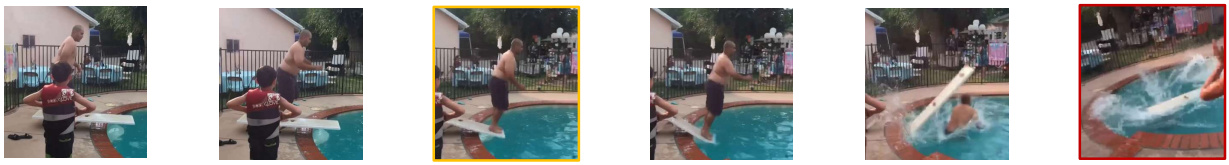
Environmental Setting: Standardized Sports Venues
Sports Category: Basketball
Macro Inducing Factors: Equipment and Facility Factors
Direct Cause Description: Unsecured basketball stand
Earliest Cue Moment (orange): It can be seen here that the basketball stand is already tilted to some extent
Most Obvious Moment (red): The basketball stand has fallen over and slammed onto the ground

Figure 2: Accident video annotation example 2.



Environmental Setting: Unstructured Human-made Spaces
Sports Category: Cycling
Macro Inducing Factors: Personal Factors
Direct Cause Description: Fall due to loss of balance
Earliest Cue Moment (orange): The cyclist is likely to lose her balance
Most Obvious Moment (red): The cyclist has tipped over onto the ground

Figure 3: Accident video annotation example 3.

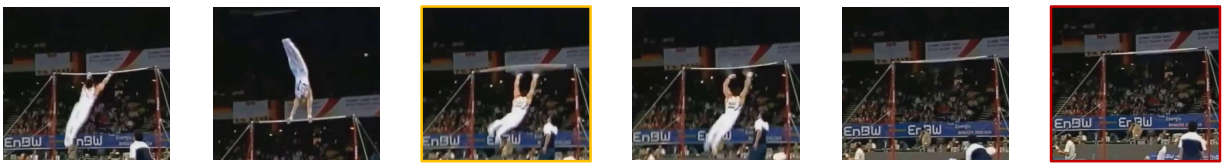


Environmental Setting: Unstructured Human-made Spaces
Sports Category: Diving
Macro Inducing Factors: Equipment and Facility Factors
Direct Cause Description: Unsecured springboard detachment
Earliest Cue Moment (orange): We can see signs of the springboard coming detached
Most Obvious Moment (red): The diver fell into the water and was hit by the springboard

Figure 4: Accident video annotation example 4.

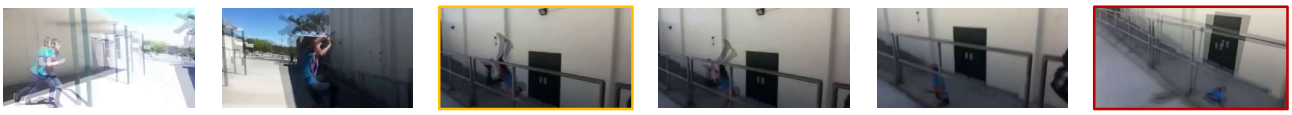
Environmental Setting: Standardized Sports Venues
Sports Category: General Fitness
Macro Inducing Factors: Equipment and Facility Factors
Direct Cause Description: Equipment Cable Breakage
Earliest Cue Moment (orange): The cable of the equipment has snapped, and the gym-goer has lost his balance
Most Obvious Moment (red): The gym-goer flipped over and hit the floor

Figure 5: Accident video annotation example 5.



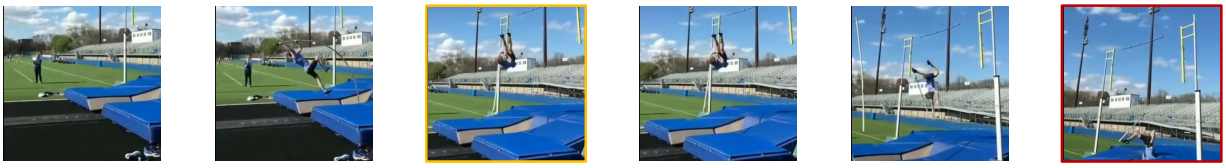
Environmental Setting: Standardized Sports Venues
Sports Category: Gymnastics
Macro Inducing Factors: Personal Factors
Direct Cause Description: Loss of grip on the horizontal bar
Earliest Cue Moment (orange): The athlete has already lost his grip on the bar at this moment
Most Obvious Moment (red): The athlete fell to the ground

Figure 6: Accident video annotation example 6.



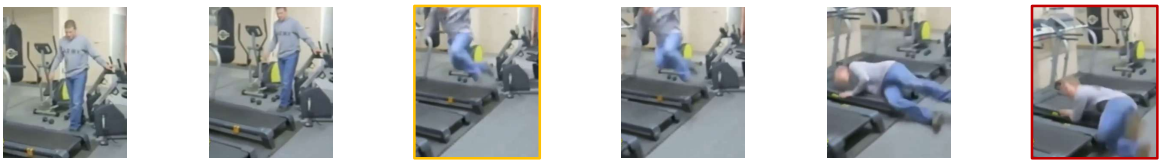
Environmental Setting: Unstructured Human-made Spaces
Sports Category: Parkour
Macro Inducing Factors: Personal Factors
Direct Cause Description: Uncontrolled fall while vaulting a railing
Earliest Cue Moment (orange): The parkour athlete has been tripped up and is about to fall
Most Obvious Moment (red): The parkour athlete fell to the ground

Figure 7: Accident video annotation example 7.



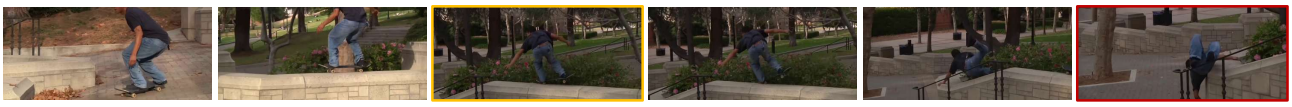
Environmental Setting: Standardized Sports Venues
Sports Category: Pole Vault
Macro Inducing Factors: Equipment and Facility Factors
Direct Cause Description: Pole breakage
Earliest Cue Moment (orange): Evidence of the pole breaking is already apparent
Most Obvious Moment (red): The athlete has already fallen onto the mat

Figure 8: Accident video annotation example 8.



Environmental Setting: Standardized Sports Venues
Sports Category: Running
Macro Inducing Factors: Personal Factors
Direct Cause Description: Stepping onto a moving treadmill
Earliest Cue Moment (orange): It can be seen that the runner has already lost his balance
Most Obvious Moment (red): The runner has fallen down

Figure 9: Accident video annotation example 9.

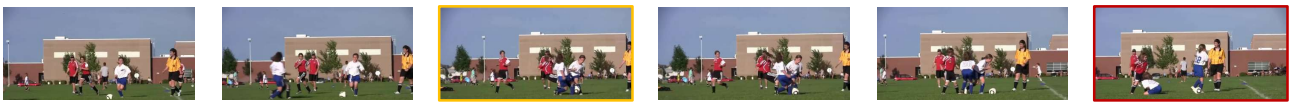


Environmental Setting: Unstructured Human-made Spaces
Sports Category: Skateboarding
Macro Inducing Factors: Personal Factors
Direct Cause Description: Fall due to loss of balance while riding
Earliest Cue Moment (orange): The skateboarder has lost his balance
Most Obvious Moment (red): The skateboarder is falling headfirst off the ledge after losing his balance

Figure 10: Accident video annotation example 10.

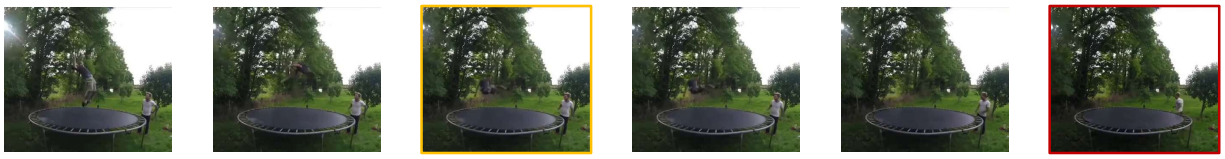
Environmental Setting: Natural & Wilderness Terrain
Sports Category: Skiing
Macro Inducing Factors: Personal Factors
Direct Cause Description: Uncontrolled jump
Earliest Cue Moment (orange): The skis planted vertically into the surface, leading to an imminent fall
Most Obvious Moment (red): The skier has hit the ground

Figure 11: Accident video annotation example 11.



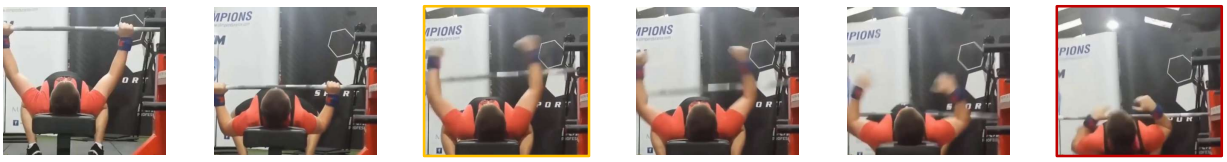
Environmental Setting: Standardized Sports Venues
Sports Category: Soccer
Macro Inducing Factors: Interpersonal Interaction Factors
Direct Cause Description: Collision with an opponent while challenging for the ball
Earliest Cue Moment (orange): Two players are about to collide
Most Obvious Moment (red): A collision has occurred, and one player has been knocked to the ground

Figure 12: Accident video annotation example 12.



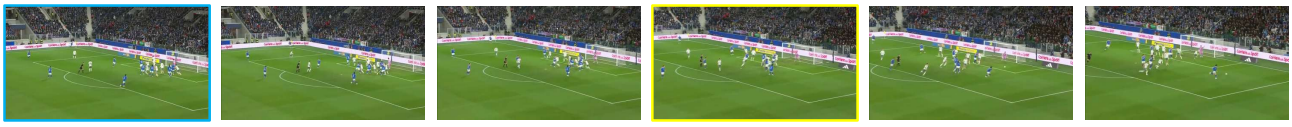
Environmental Setting: Unstructured Human-made Spaces
Sports Category: Trampolining
Macro Inducing Factors: Personal Factors, Equipment and Facility Factors
Direct Cause Description: Thrown off the trampoline without a safety net
Earliest Cue Moment (orange): The person is launched into the air and is about to fly off the trampoline
Most Obvious Moment (red): The person has been thrown off the trampoline and hit the ground

Figure 13: Accident video annotation example 13.



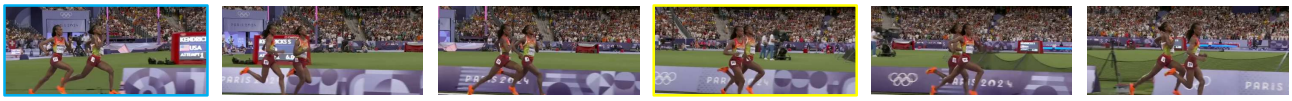
Environmental Setting: Standardized Sports Venues
Sports Category: Weightlifting
Macro Inducing Factors: Personal Factors
Direct Cause Description: Losing grip on the barbell during bench press
Earliest Cue Moment (orange): The lifter has lost control of the barbell
Most Obvious Moment (red): The barbell has fallen onto the lifter

Figure 14: Accident video annotation example 14.



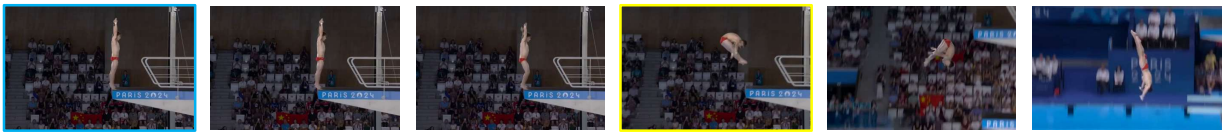
Environmental Setting: Standardized Sports Venues
Sports Category: Soccer
First Frame of the Video (blue): The players are getting ready to battle for the ball
High-Amplitude Movement Moment (yellow): The players are battling for the ball

Figure 15: Safety video annotation example 1.



Environmental Setting: Standardized Sports Venues
Sports Category: Running
First Frame of the Video (blue): Two athletes are running
High-Amplitude Movement Moment (yellow): One athlete is about to overtake another

Figure 16: Safety video annotation example 2.



Environmental Setting: Standardized Sports Venues
Sports Category: Diving
First Frame of the Video (blue): The diver is getting ready to take off from the platform
High-Amplitude Movement Moment (yellow): The diver takes off into the air

Figure 17: Safety video annotation example 3.

E Distributions of Dataset Attributes

In this section, we detail the distributions of Environmental Settings, Sports Categories, and Macro Inducing Factors across our dataset. Specifically, Table 1 presents the frequencies of all three attributes within the accident dataset, while Table 2 details the distributions of only Environmental Settings and Sports Categories within the safety dataset.

Table 1: Attribute frequencies within the accident dataset.

Category	Count
Environmental Settings	
Standardized Sports Venues	1223
Unstructured Human-made Spaces	794
Natural and Wilderness Terrain	423
Sports Categories	
Skiing	248
Trampolining	234
General Fitness	231
Weightlifting	227
Cycling	225
Parkour	220
Basketball	172
Running	164
Baseball	156
Soccer	129
Diving	125
Gymnastics	121
Skateboarding	108
Pole Vault	80
Macro Inducing Factors	
Personal Factors	1634
Equipment and Facility Factors	557
Interpersonal Interaction Factors	453
Venue and Environmental Factors	396

Table 2: Attribute frequencies within the safety dataset.

Category	Count
Environmental Settings	
Standardized Sports Venues	361
Unstructured Human-made Spaces	73
Natural and Wilderness Terrain	14
Sports Categories	
Diving	32
Soccer	32
Skateboarding	32
Basketball	32
Pole Vault	32
Running	32
Weightlifting	32
Skiing	32
Trampolining	32
Cycling	32
General Fitness	32
Gymnastics	32
Baseball	32
Parkour	32

F Detailed Experimental Results

In this section, we present the detailed experimental results. Specifically, Table 3 details the results of the Full-Video Evaluation, Table 4 reports the results of the Temporal Window Evaluation, and Table 5 presents the Diagnostic Analysis. Please note that in the Diagnostic Analysis results, higher values in the table denote a higher false alarm rate, where safe videos are incorrectly identified as unsafe.

Table 3: Results of the Full-Video Evaluation. Here, P^1 denotes $Prompt_1^1$, P^2 denotes $Prompt_1^2$, and P^3 denotes $Prompt_1^3$.

Model	Prompt	Scores		
		D_1	D_2	D_3
Open-source Models				
InternVL3.5-8B	P^1	0.46	0.34	0.10
	P^2	0.59	0.46	0.21
	P^3	0.86	0.67	0.19
Qwen3-VL-8B	P^1	0.47	0.38	0.13
	P^2	0.47	0.37	0.14
	P^3	0.89	0.80	0.24
Proprietary Models				
Doubao-seed-1.8	P^1	0.97	0.88	0.42
	P^2	0.97	0.88	0.53
	P^3	1.00	0.88	0.64
GPT-4o	P^1	0.78	0.69	0.33
	P^2	0.80	0.70	0.38
	P^3	0.95	0.90	0.46
GPT-5	P^1	0.92	0.86	0.44
	P^2	0.94	0.88	0.65
	P^3	0.98	0.90	0.64
Gemini-3-Flash	P^1	0.81	0.75	0.38
	P^2	0.77	0.70	0.42
	P^3	0.99	0.91	0.65
Gemini-3-Pro	P^1	0.85	0.77	0.45
	P^2	0.82	0.64	0.26
	P^3	0.98	0.85	0.61

G Preliminary Exploration of Mitigation Strategy

G.1 Motivation and Method

To mitigate the over-warning bias induced by hazard-seeking prompts without requiring fine-tuning, we introduce a "Proposer-Verifier" baseline. The pipeline consists of two stages:

- **Proposer:** The MLLM predicts potential hazards using $Prompt_2^1$.
- **Verifier:** If a hazard is reported, the MLLM acts as a "skeptic" to re-evaluate the Proposer's prediction alongside the original video frames, determining whether a legitimate hazard is truly developing.

G.2 Results and Discussion

We evaluate this baseline on the `Doubao-seed-1.8` model during the earliest video phase (Window A). We report the D_1 scores on both the accident dataset and the safety dataset.

As shown in Table 6, the dual-agent strategy achieves a clear trade-off. Compared to P^1, it successfully reduces the false alarm rate on the Safety Dataset from 0.88 to 0.26. However, the hazard detection sensitivity on the Accident Dataset also drops from 0.88 to 0.71. Furthermore, its false alarm rate (0.26) remains double that of the neutral prompt P^2 (0.13), indicating that the "skeptic" agent is still partially influenced by the Proposer's initial bias.

This sub-optimal performance demonstrates that simple heuristic prompt chaining is insufficient to bridge the capability gap in early hazard anticipation. This further underscores the challenging nature of our proposed dataset and the necessity for future architectural innovations.

H Annotator Details and Quality Control

Data annotation for this study was conducted collaboratively by members of our internal research team. All annotators possess good English proficiency and, as sports enthusiasts, have relevant domain background knowledge.

This study involves multiple rounds of annotation tasks. Before each official round, the research team developed detailed annotation guidelines and provided 25 video examples for preliminary training. To ensure consistency in annotation standards, thorough internal discussions were held to guarantee that all annotators accurately understood the requirements. Finally, all annotated samples underwent rigorous review by the research team to ensure data quality.

I Limitations

Although this study provides an analysis of the vulnerabilities of Multimodal Large Language Models in the domain of proactive safety, several limitations remain:

(1) **Absence of the Audio Modality:** Since the raw data was collected from online video platforms, a significant number of videos either lack original soundtracks, have been muted, or contain post-edited background music. Consequently, the benchmark in this study focuses exclusively on the interaction between visual and textual modalities, failing to incorporate audio features (e.g., the sound of objects breaking or severe collisions, which serve as crucial physical warning signals) into the evaluation of the models' proactive warning capabilities.
(2) **Limited Scope of Evaluated Models:** The evaluation in this study primarily focuses on general-purpose MLLMs, without including specialized video anomaly detection models or action prediction models for comparison.
(3) **Limited Scenario Coverage:** SPA-Bench currently focuses predominantly on sports scenarios. While sports provide an excellent testbed for highly dynamic and complex physical interactions, real-world proactive safety also encompasses numerous domains such as autonomous driving, industrial manufacturing, and daily caregiving. Future work needs to further expand the scenario diversity of the dataset.

Table 4: Results of the Temporal Window Evaluation. Here, P^1 denotes $Prompt_2^1$, and P^2 denotes $Prompt_2^2$.

Model	Prompt	Window A			Window B			Window C		
		D_1	D_2	D_3	D_1	D_2	D_3	D_1	D_2	D_3
Open-source Models										
InternVL3.5-8B	P^1	0.18	0.14	0.04	0.39	0.31	0.15	0.47	0.38	0.22
	P^2	0.05	0.04	0.02	0.26	0.20	0.12	0.42	0.34	0.21
Qwen3-VL-8B	P^1	0.41	0.36	0.10	0.58	0.52	0.20	0.66	0.60	0.29
	P^2	0.09	0.08	0.02	0.24	0.20	0.08	0.35	0.31	0.16
Proprietary Models										
Doubao-seed-1.8	P^1	0.88	0.65	0.40	0.93	0.76	0.54	0.95	0.79	0.61
	P^2	0.59	0.47	0.27	0.86	0.74	0.51	0.93	0.81	0.60
GPT-4o	P^1	0.42	0.37	0.15	0.65	0.59	0.33	0.72	0.67	0.43
	P^2	0.24	0.22	0.08	0.59	0.54	0.30	0.73	0.67	0.43
GPT-5	P^1	0.67	0.56	0.33	0.81	0.73	0.53	0.85	0.78	0.63
	P^2	0.64	0.60	0.27	0.87	0.82	0.54	0.92	0.88	0.67
Gemini-3-Flash	P^1	0.75	0.57	0.33	0.91	0.76	0.57	0.93	0.81	0.63
	P^2	0.41	0.37	0.20	0.74	0.68	0.45	0.82	0.77	0.57
Gemini-3-Pro	P^1	0.78	0.57	0.35	0.93	0.78	0.58	0.95	0.82	0.65
	P^2	0.42	0.38	0.21	0.76	0.69	0.47	0.85	0.78	0.60

Table 5: Results of the Diagnostic Analysis. Here, P^1 denotes $Prompt_2^1$, P^2 denotes $Prompt_2^2$, and Gap represents the performance difference between P^1 and P^2 on D_1.

Model	Temporal Truncation Test			Static First-Frame Test		
	P^1	P^2	Gap	P^1	P^2	Gap
Open-source Models						
InternVL3.5-8B	0.16	0.01	0.15	0.12	0.02	0.10
Qwen3-VL-8B	0.17	0.04	0.13	0.18	0.10	0.08
Proprietary Models						
Doubao-seed-1.8	0.88	0.13	0.75	0.67	0.13	0.54
GPT-4o	0.32	0.10	0.22	0.20	0.08	0.12
GPT-5	0.59	0.28	0.31	0.48	0.25	0.23
Gemini-3-Flash	0.73	0.12	0.61	0.72	0.11	0.61
Gemini-3-Pro	0.48	0.12	0.36	0.42	0.14	0.28

Table 6: Results of the Mitigation Strategy. An ideal method should maintain high detection on the accident dataset while minimizing false alarms on the safety dataset. Here, P^1 denotes $Prompt_2^1$, and P^2 denotes $Prompt_2^2$.

Model	Method	D_1 Window A	
		Accident Dataset	Safety Dataset
Doubao-seed-1.8	P^1	0.88	0.88
	P^2	0.59	0.13
	P^1 + Verifier	0.71	0.26

(4) **Lack of Structured Annotation for Direct Causes of Accidents (H_2):** This study employs natural language to describe the direct causes of accidents (H_2) and has not yet constructed a structured classification label system. Such unformatted, free-text annotation has inherent limitations. First, annotators may use varying linguistic expressions for the same event, easily introducing semantic divergence. Second, constrained by the annotators' prior knowledge, natural language descriptions may suffer from information omission. Furthermore, free text often lacks sufficient professional rigor at the kinematic and biomechanical levels. These factors inevitably introduce subjective noise, thereby limiting the precision and objectivity in evaluating the models' fine-grained hazard tracing capabilities (the D_3 dimension) to some extent.